Universidad del Desarrollo

POLICICS

Networks Decide for Us

The Emergence of Adoption in Homophily-Imputed Social Topologies

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The question

- Collective behaviors — innovations, mobilizations — spread as **adoption cascades** on social networks.
- We can measure **who people are** (preferences, from surveys), and we know **how they cluster** (multidimensional homophily) [1,2].
- Models usually run on synthetic networks, with invented preferences.
- Our experiment: **fix the real people**, and rewire the ties.

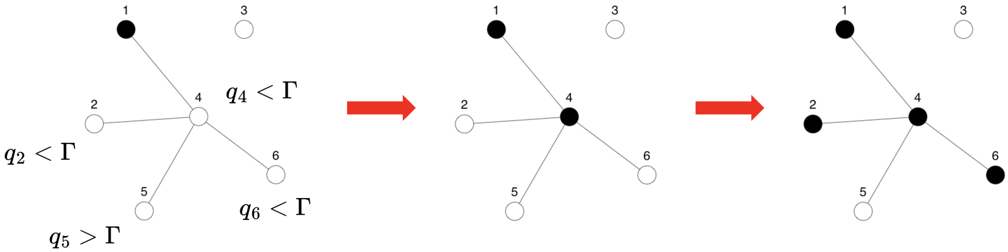
Conditional on *fixed individual preferences*, to what extent does the **empirical network structure** — shaped by homophily — govern the diffusion of collective behaviors?

The model

Adoption, part 1 — Rational Choice

- A behavior has an *Intrinsic Utility Level* **IUL**, $\Gamma \in [0, 1]$.
- Each person has a *Minimum Utility Requirement* **MUR**, $q_i \in [0, 1]$, representing their aversion to adopting the behavior.

An agent adopts on its own iff $\boxed{q_i \leq \Gamma \implies a_i = 1}$

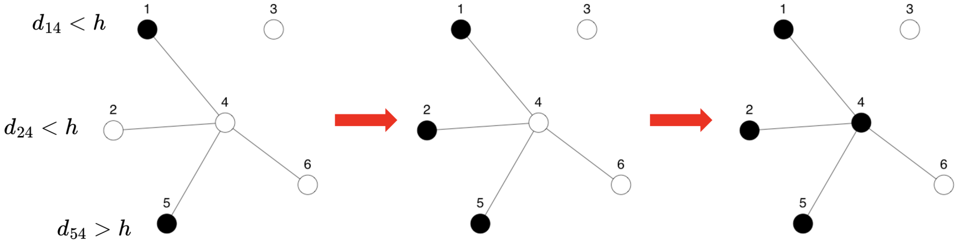


Adoption, part 2 — Selective Social Influence

Adopt by peer contagion if $\tau_i \le \tilde{E}_i$ [3] — but only those socially *close* count [4]:

$$\tilde{E}_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij}\, \tilde{a}_j}{\sum_{j \neq i} \mathbf{X}_{ij}}, \qquad \tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \le \sigma(MSP - d_{ij})$$

- There is a Maximum Social Proximity **MSP** where the limit of influence sits.



Imputing structure & running the model

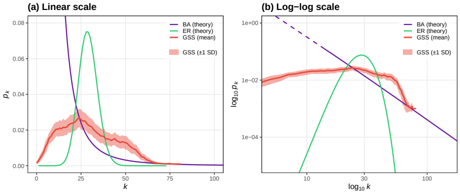
Imputing a network onto a survey

Homophily strengths $\hat{\boldsymbol{\beta}}$ [5,6] → Construct realistic networks → Add propensity scores q_i to each node

Tie probability is a function of social distance (age, sex, education, race, religion):

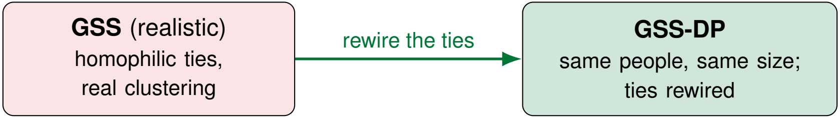
$$\text{logit}\ \Pr(Y_{ij} = 1) = \alpha + \boldsymbol{\beta}^\top \mathbf{d}_{ij}$$

$\hat{\boldsymbol{\beta}}$ comes from GSS confidant data [5]; propensities q_i come from real survey answers (innovation, collective action).



Right-skewed degree structure that Erdös-Rényi / Barabási-Albert models fail to match jointly.

The experiment — fix the people, rewire the ties



- Random rewiring creates **long-range bridges** — classic theory says this should *help* diffusion [8].
- Or, if adoption were just the sum of individual preferences, rewiring the ties should do nothing...

Comprehensive simulation design

Hybrid social contagion via `netdiffuseR` [7], over plausible **GSS** vs. **GSS-DP** networks.

- **IUL** (Γ): 41 levels $[0, 1]$. **MSP** (h): 13 levels $[0, 1]$.
- **Thresholds** $\tau_i \sim \mathcal{N}(\mu, \sigma)$: $\mu \in \{.3, .4, .5, .6\}$, $\sigma \in \{.08, .12, .16, .20\}$.
- **Seeding**: random, central, marginal, closeness, eigenvector.

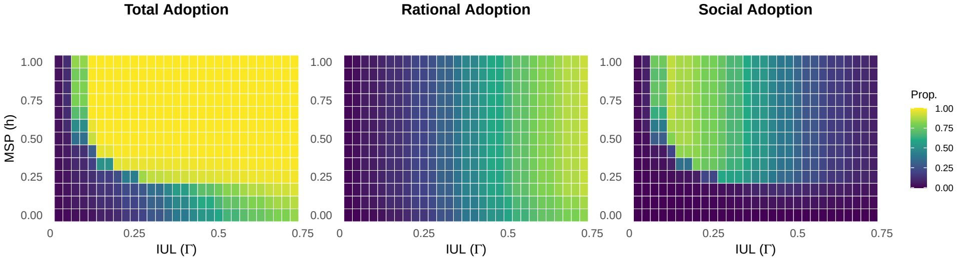
Total: $41 \times 13 \times 4 \times 4 \times 5 \times$ (runs) $\approx$ 4×10^6 **simulations**

We then fit *Big Additive Models* (BAM) to isolate the structural effect:

$$\text{logit}\left[\mathbb{E}(\Phi)\right] = \beta_0 + \mathbf{Z}\boldsymbol{\beta} + \beta_{\text{R}}\, \mathbb{I}_{\text{Randomized}} + s(\text{IUL}, \text{MSP})$$

Results

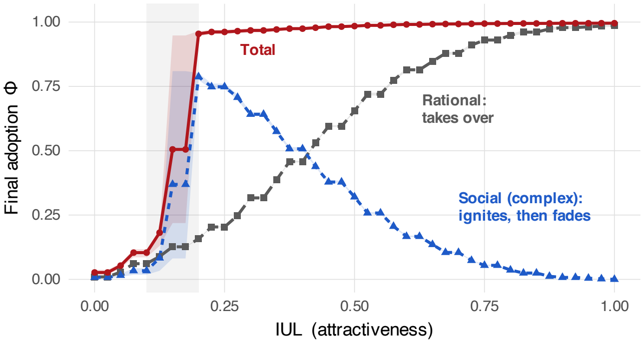
Result 1 — The dynamics



Mean adoption over the attractiveness (IUL) – openness (MSP) plane.

Rational adoption grows with attractiveness; **social** adoption is more complex, so total adoption stays high over a wide region *even when the behavior is barely attractive*.

Result 2 — A *non-structural* blocking of complex contagion



Centola & Macy [9] already showed a structural barrier to complex diffusion.
This is a **non-structural** blocking of diffusion of collective behavior.

Result 3 — What happens when we rewire?

Same individuals, same preferences.
And the **rewiring created long-range bridges** that should have helped [8,9].

- even when we have both Simple and Complex contagion, the bridges are useless.

Homophilic **echo chambers** give social reinforcement and redundancy.

GSS vs GSS-DP	β	OR	odds
Total adoption	-0.850^{***}	0.43	-57%
Social adoption	-0.426^{***}	0.65	-35%
Rational adoption	-0.114^{***}	0.89	-11%

$^{***}p < 0.001$; BAM, ref. = GSS; controls: thresholds, seeding, $s(\mathrm{IUL}, \mathrm{MSP})$. Dev. expl. $\approx 83\%$.

Conclusions

1. **A method:** build **realistic** whole-networks for survey respondents out of **empirical homophily**.

2. **Three findings:**

    1. Social influence can explain adoption even for *low attractive* behaviors.
    2. A **non-structural** blocking of diffusion of collective behavior happens because of the Social Influence channel.
    3. Rewiring the ties cuts the **odds of adoption by** -57%, contrary to expectations.

Where a behavior ends up is not the sum of individual preferences
it is **governed by the structure of our real ties.**

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Thanks!

Scan: preprint draft (Zenodo)

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Backup

Backup — Which ingredient? Scrambling *people*, not ties

A third network: **GSS-SH** — keep the graph *byte-identical* (every triangle, every clique), and **shuffle who sits where**.

Your neighbors stay your neighbors in the graph — but they become *social strangers*: mean social distance on ties jumps $+74\%$.

vs. GSS	OR	odds drop
GSS-SH (scramble people)	0.465	-53%
Randomized (rewire ties)	0.432	-57%

Scrambling the people alone reproduces $\sim$**91%** of the full penalty: **neighbor similarity** carries the premium.

Why not a clean "topology-only" cell? Because in these networks **clustering emerges from homophily** — re-simulating the homophily-only ERGM regenerates the triangles. They come as a package: *clustering is crystallized homophily.*

(3-level BAM; $\beta_{\text{SH}} = -0.766$, $\beta_{\text{R}} = -0.840$; main-slide 2-level estimates differ negligibly.)

Backup — Mapping social adoption to a ferromagnet

Ising system		**NDFU**
Spin $s_i \in \{0,1\}$	$\leftrightarrow$	Adoption state a_i
Magnetization M	$\leftrightarrow$	Adoption $\Phi = N_{\text{ad}}/N$
External field H	$\leftrightarrow$	**IUL**
Temperature T	$\leftrightarrow$	**MSP**
Exchange J_{ij}	$\leftrightarrow$	Clustering w_{ij}
Susceptibility χ	$\leftrightarrow$	$\mathrm{Var}(\Phi)$

Pseudo-Hamiltonian [12]:

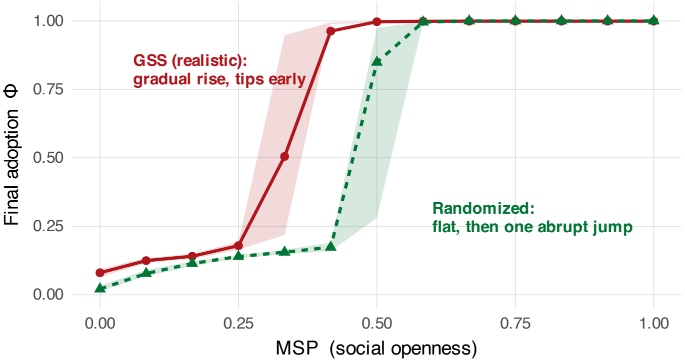
$$\mathcal{H} \approx -\sum_{\langle i,j \rangle} w_{ij}(h)\, s_i s_j \; - \; \Gamma \sum_i s_i$$

Low MSP $\rightarrow$ rigid echo chambers.

High MSP $\rightarrow$ boundaries “melt”.

This is why MSP behaves as a social “temperature”.

Backup — The tipping point, compared



Rewiring the ties also changes the *shape* of the tipping: the realistic network tips **earlier and gradually**, through intermediate states;
the randomized one stays flat and then jumps in **one abrupt step** — the hit-or-flop pattern of Tur et al. [10].

Backup — The shape of change, quantified

Near the tipping point, the variance of outcomes diverges as a power law, $\chi = \mathrm{Var}(\Phi) \sim |MSP - MSP_c|^{-\gamma}$:

Susceptibility exponent γ (mean)	**GSS**	**GSS-SH**	**Randomized**
	≈ 0.91	≈ 1.08	≈ 3.9

- GSS and GSS-SH: clean power law, $\gamma \approx 1$ — a **continuous**, signposted transition (rising variance before the tip).
- Randomized: no clean power law — the fit "$\gamma \approx 4$" is the fingerprint of a **discontinuous**, all-or-nothing jump.
- These are ensemble statements (distribution over comparable runs), not single-history forecasts.

Backup — Technical details

- **Social distance** $d_{ij} \in [0,1]$ — Gower distance over the Blau-space attributes (race, sex, religion, age, education):

$$d_{ij} = \frac{1}{p}\sum_{k=1}^{p} \delta_{ij}^{k}, \qquad \delta_{ij}^{k} = \begin{cases} \mathbb{1}\left[x_i^k \neq x_j^k\right] & \text{categorical (race, sex, religion)} \\ \dfrac{\left|x_i^k - x_j^k\right|}{\max_k - \min_k} & \text{numeric (age, education)} \end{cases}$$

* The idiosyncrasy that drives adoption of barely-attractive behaviors here is **measured preference** and **realistic structure**, not stochastic noise (Macy & Evtushenko [11]).